



Namal University Mianwali

Department of Computer Science

Course Title:	Parallel and Distributed Computing
Course Code:	CS-440
Credits (Theory + Lab):	3 (3+0)
Lecture Hours/Week:	3
Pre-Requisite:	CS-330 Operating Systems
Brief Course Description: This course introduces the principles of parallel and distributed computing, covering key concepts, architectures, and programming models. Students will learn about speedup, scalability, Amdahl's Law, and different architectures such as multiprocessors, shared memory, distributed memory systems (clusters and grids), and GPU/heterogeneous systems. The course covers parallel programming paradigms including threads, message passing (MPI), OpenMP, and accelerator programming (CUDA/OpenCL). Students will explore parallel algorithms, concurrency, synchronization, load balancing, and performance optimization. Through practical projects and case studies, they will gain hands-on experience in developing scalable and efficient parallel and distributed systems for real-world applications.	

Name:	Dr. Muzamil Ahmed
Lecture Timings:	Tuesday (2:00 PM to 3:30 PM) and Wednesday (9:00 AM to 10:30 AM),
Office Hours:	Monday & Thursday (11:00 AM to 2:00 PM)
Office Location:	Office 6
Office Phone:	Ext. 184
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Unit No.	Topics	No of Teaching Hours
1	Introduction to Parallel and Distributed Systems	7.5
2	Multiprocessors and Shared Memory Systems	9
3	Software Architecture for Parallel Systems	6
4	Parallel Algorithms	7.5
5	Cloud, GPU & Parallel Programming	9
6	Concurrency, Synchronization, and Load Balancing	9
Total Contact Hours		48

Sr. No	Unit No	CLO Statement	Taxonomy Level	GA
CLO-1	1	Explain the fundamental concepts of parallel and distributed computing	C2	1
CLO-2	2,5	Apply parallel and distributed computing techniques to solve computational problems	C3	2
CLO-3	4,6	Analyze the performance of parallel and distributed systems	C4	3
CLO-4	3,4,6	Utilize appropriate programming models and tools for developing parallel and distributed systems	C3	5

Lecture #	Unit #	Topics Covered in Class	Reference in Book / Assessments
1	0	Introduction to the Course - Course Overview, Syllabus, and Learning Objectives - Grading Policy (Quizzes, Assignments, Midterm, Final Exam, Projects)	Reading Assignment
2	1	Real-World Environments that Exhibit Parallelism - Overview of Parallelism in Various Fields: Scientific Computing, Graphics, AI	Pacheco: Ch 1
3	1	Module 1: Introduction to Parallel and Distributed Systems Introduction to Parallel and Distributed Systems - Why use Parallel and Distributed Computing?	Pacheco: Ch 1
4	1	Why not use Parallel and Distributed Computing? - Understanding the Challenges, Overhead, and Limitations	Pacheco: Ch 1
5	1	Speedup and Scalability - Concepts of Speedup, Superlinear Speedup, Scalability Issues	Pacheco: Ch 2
6	1	Amdahl's Law - Impact on Performance, Understanding the Serial and Parallel Portions	Pacheco: Ch 2 Quiz 1
7	1	Hardware Architecture for Parallel Computing - Multi-core Processors, GPUs, and Heterogeneous Systems	Pacheco: Ch 2
8	2	Module 2: Multiprocessors and Shared Memory Systems Multiprocessors - Symmetric and Asymmetric Multiprocessing	Pacheco: Ch 4
9	2	Shared Memory Architectures - Cache Coherence, Memory Consistency Models	Pacheco: Ch 4 Assignment 1
10	2	Networks of Workstations (NOW) - Clusters and Networked Systems	Pacheco: Ch 4
11	2	Distributed Memory Systems - Key Differences, Advantages, and Drawbacks	Pacheco: Ch 3
12	2	Computer Clusters - High-Performance Clustering and Use Cases	Pacheco: Ch 3
13	3	Module 3: Software Architecture for Parallel Systems Threads and Shared Memory Programming Paradigm - Overview of Threading Models and Shared Memory Models	Pacheco: Ch 2 Quiz 2
14	3	Processes and Message Passing Programming Paradigm - Communication between Processes using MPI	Pacheco: Ch 3

15	3	Processes and Message Passing Programming Paradigm (cont.) - Communication between Processes using MPI	Pacheco: Ch 3
16	3	Distributed Shared Memory (DSM) Concepts and Implementation, Example Architectures	Pacheco: Ch 3
Mid Term Exam			
17	4	Module 4: Parallel Algorithms Divide and Conquer Algorithms - Parallel Approaches for Divide and Conquer Problems	Pacheco: Ch 5 Semester Project
18	4	Greedy Methods in Parallel Algorithms - Examples of Greedy Algorithms with Parallel Approaches	Pacheco: Ch 5
19	4	Dynamic Programming and Parallelism - Parallel Dynamic Programming Approaches and Examples	Pacheco: Ch 5
20	4	Backtracking in Parallel - Parallel Techniques in Backtracking Algorithms	Pacheco: Ch 5 Quiz 3
21	5	Module 5: Cloud, GPU & Parallel Programming Introduction to Cloud Computing - Overview of Cloud Systems, Services, and Use Cases	L. Warner: Ch 1
22	5	Virtualization and Cloud Services (IaaS, PaaS, SaaS) Virtualization Concepts, Use of Hypervisors, Cloud Services Models	L. Warner: Ch 5
23	5	GPU Architectures & CUDA/OpenCL Basics Parallelism on Accelerators	Hennessy/Patterson: Ch 15
24	5	Heterogeneous Computing Models - CPU and GPU Collaboration	Hennessy/Patterson: Ch 15 Assignment 2
25	5	OpenMP Basics - Directives for Shared Memory Parallelism	Pacheco: Ch 6
26	5	OpenMP Programming Practice - Loops, Sections, Synchronization	Pacheco: Ch 6
27	6	Module 6: Concurrency, Synchronization, and Load Balancing Concurrency and Synchronization in Parallel Computing - Critical Sections and Deadlocks	Pacheco: Ch 5 Quiz 4
28	6	Data and Work Partitioning in Parallel Computing - Partitioning Techniques, Task Decomposition, Data Decomposition	Pacheco: Ch 5
29	6	Recursive Decomposition and Parallelization Strategies - Recursive Problems in Parallel Environments, Granularity Issues	Pacheco: Ch 5
30	6	Load Balancing in Parallel Systems - Static vs Dynamic Load Balancing, Strategies, and Tools	Pacheco: Ch 5
31	6	Project Presentations – I (Group Project)	
32		Project Presentations – II (Group Project)	
Final Term Exam			

Assessment Tool	CLO1	CLO2	CLO3	CLO4
Quizzes	✓	✓	✓	
Assignments		✓	✓	✓
Projects	✓	✓	✓	✓
Mid Term Exam	✓	✓	✓	
Final Term Exam	✓	✓	✓	✓

Textbooks:

1. An Introduction to Parallel Programming by Peter S. Pacheco, 2nd Edition.
2. Computer Architecture: A Quantitative Approach by John L. Hennessy & David A. Patterson, 6th Edition

Reference Material:

3. Designing and Building Parallel Programs, by Ian Foster
4. Distributed Systems, Third edition, by Maarten van Steen, Andrew S. Tanenbaum
5. Microsoft Azure For Dummies 1st Edition by Timothy L. Warner 2020

- **Attendance Policy:** Students must attend 80% of this course's lectures. Students who fall short of the required attendance percentage cannot appear in the terminal examination.
- **Plagiarism Policy:** Namal University's rules and regulations prohibit plagiarism, copying, and other dishonest behavior. Violators will face serious disciplinary action.
- **Course Assessment:**

Assessment Tool	Quiz	Assignment	Project/Tools	Mid Term Exam	Final Term Exam	Total
Theory (T)	15	05	10	30	40	100

(i) I verify that the content of this document is correct and up-to-date.

<u>Dr. Muzamil Ahmed</u> Instructor's Name and Signature	<u>19-08-2025</u> Date
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(ii) I have reviewed course-outline and state that it complies with Namal University policies and guidelines.

<u>Dr. Muhammad Ali Shahid</u> Name and Signature of Head of Department	<u>19-08-2025</u> Date
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